Conditional Probability and Bayes Rule

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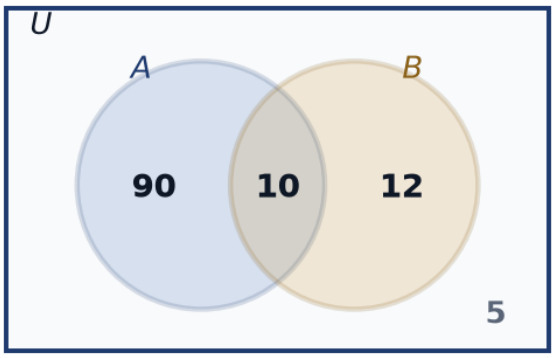
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.05$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

2 [Foundational] • Target: Joint Probability and Independence

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.10$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.



3 [Foundational] • Target: Conditional Probability and the Chain Rule

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.15$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

4 [Foundational] • Target: Law of Total Probability

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.20$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

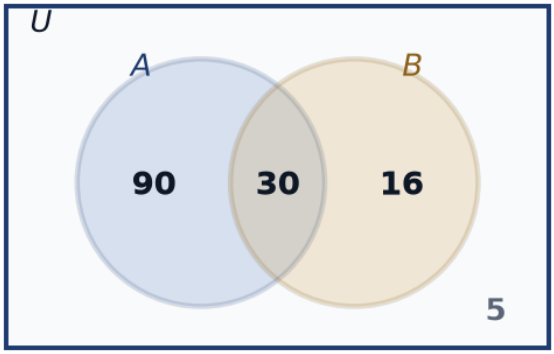
5 [Foundational] • Target: Bayes Theorem in Machine Learning and Screening

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.25$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

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6 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.30$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.



7 [Foundational] • Target: Joint Probability and Independence

In anomaly detection, anomaly prior $P(A) = 0.02$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Conditional Probability and the Chain Rule

In anomaly detection, anomaly prior $P(A) = 0.04$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

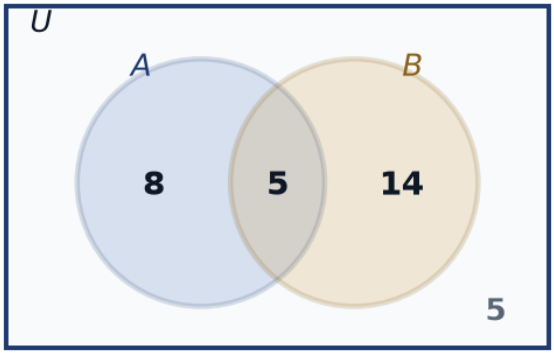
9 [Intermediate] • Target: Law of Total Probability

In anomaly detection, anomaly prior $P(A) = 0.06$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

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10 [Intermediate] • Target: Bayes Theorem in Machine Learning and Screening

In anomaly detection, anomaly prior $P(A) = 0.08$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.



11 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In anomaly detection, anomaly prior $P(A) = 0.10$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

12 [Intermediate] • Target: Joint Probability and Independence

In anomaly detection, anomaly prior $P(A) = 0.12$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly? Show your complete algebraic work.

13 [Intermediate] • Target: Conditional Probability and the Chain Rule

Consider a medical diagnostic test for a rare disease. The prevalence of the disease in the general population is $P(D) = 0.001$. The test has a sensitivity of $P(+|D) = 0.99$ (the probability of a positive test given the person has the disease) and a specificity of $P(-|egD) = 0.98$ (the probability of a negative test given the person does not have the disease). If a randomly selected individual tests positive, what is the probability that they actually have the disease, given that we are interested in the positive predictive value (PPV) of the test, and the prior probability of having the disease is adjusted by a factor f such that the new prior is $P'(D) = f \cdot P(D)$? For this specific scenario, suppose $f = 1.5$ and we want to find $P'(D|+)$.

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14 **[Intermediate]** • **Target: Law of Total Probability**

A quality control engineer is monitoring the number of defects produced by two different machines, Machine A and Machine B. Let X be the number of defects produced by Machine A in a given hour, and Y be the number of defects produced by Machine B in the same hour. The joint probability distribution of X and Y is given in the table below: | $P(X = x, Y = y)$ | $y = 0$ | $y = 1$ | $y = 2$ | | :----------: | :-------: | :-------: | :-------: | | $x = 0$ | 0.10 | 0.15 | 0.05 | | $x = 1$ | 0.20 | 0.30 | 0.10 | | $x = 2$ | 0.05 | 0.05 | 0.00 | What is the expected value of the total number of defects produced by both machines in a given hour, $E(X + Y)$? Show your complete algebraic work.

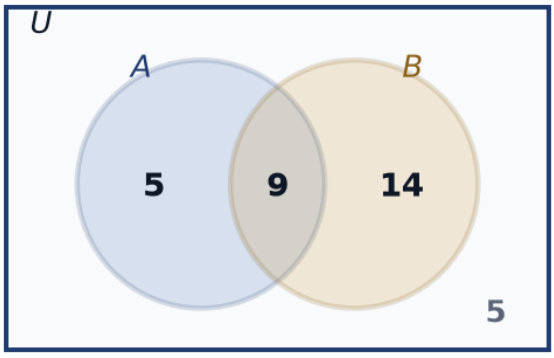
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Bayes Theorem in Machine Learning and Screening** • **Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

A medical diagnostic test for a rare disease is designed to be highly sensitive and specific. The disease affects 1 in every 1000 individuals in the population. The test has a sensitivity of 99% (meaning it correctly identifies 99% of individuals who have the disease) and a specificity of 95% (meaning it correctly identifies 95% of individuals who do not have the disease). If a randomly selected individual from the population tests positive, what is the probability that they actually have the disease? Show your complete algebraic work.

16 **[Challenging]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises** • **Cross-Lesson: 6.01 Linear Regression and OLS**

A medical test for a rare disease has a 99\% accuracy rate in detecting the disease when it is present (True Positive Rate) and a 95\% accuracy rate in identifying individuals without the disease (True Negative Rate). If the prevalence of the disease in the population is 0.5\%, what is the probability that a person who tests positive actually has the disease? Show your complete algebraic work.



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17 [Challenging] • Target: Joint Probability and Independence • Cross-Lesson: 7.01 Logistic Regression and Classification

Consider two events, A and B, with the following probabilities: $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. What is the conditional probability of A given B, denoted as $P(A|B)$? Show your complete algebraic work.

18 [Challenging] • Target: Conditional Probability and the Chain Rule • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.10$. What is the probability of the complement class $P(egC_1)$? Show your complete algebraic work.

19 [Challenging] • Target: Law of Total Probability • Cross-Lesson: 1.01 Vectors and Embeddings in AI

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.20$. What is the probability of the complement class $P(egC_1)$? Show your complete algebraic work.

20 [Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.60$. What is the probability of the complement class $P(egC_1)$? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class \(C_1 \) is \(P(C_1) = 0.10 \). What is the probability of the complement class \(P(\neg C_1) \)?

a. \(0.90 \)

b. \(0.95 \)

c. \(0.05 \)

d. \(1.10 \)

22 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class \(C_1 \) is \(P(C_1) = 0.20 \). What is the probability of the complement class \(P(\neg C_1) \)?

a. \(0.80 \)

b. \(0.90 \)

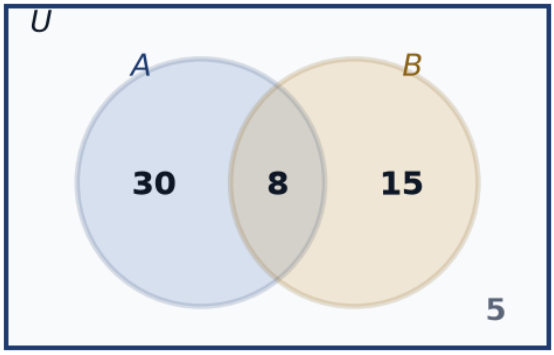
c. \(0.10 \)

d. \(1.20 \)

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23 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class \(C_1 \) is \(P(C_1) = 0.30 \). What is the probability of the complement class \(P(\neg C_1) \)?



a. \(0.70 \)

b. \(0.85 \)

c. \(0.15 \)

d. \(1.30 \)

24 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class \(C_1 \) is \(P(C_1) = 0.40 \). What is the probability of the complement class \(P(\neg C_1) \)?

a. \(0.60 \)

b. \(0.80 \)

c. \(0.20 \)

d. \(1.40 \)

25 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class \(C_1 \) is \(P(C_1) = 0.50 \). What is the probability of the complement class \(P(\neg C_1) \)?

a. \(0.50 \)

b. \(0.75 \)

c. \(0.25 \)

d. \(1.50 \)

26 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

In a dataset, the prior probability of class \(C_1 \) is \(P(C_1) = 0.60 \). What is the probability of the complement class \(P(\neg C_1) \)?

a. \(0.40 \)

b. \(0.70 \)

c. \(0.30 \)

d. \(1.60 \)

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27 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A machine learning diagnostic test has sensitivity $P(\text{Positive} \mid \text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.05$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

a. 0.0450

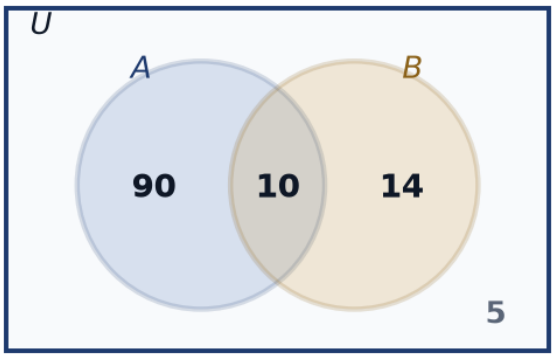
b. 0.9500

c. 0.8500

d. 0.0050

28 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A machine learning diagnostic test has sensitivity $P(\text{Positive} \mid \text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.10$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.



a. 0.0900

b. 1.0000

c. 0.8000

d. 0.0100

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Joint Probability and Independence

If two events A and B are statistically independent with $P(A) = 0.4$ and $P(B) = 0.3$, find the joint probability $P(A \cap B)$.

a. 0.12

b. 0.70

c. 0.10

d. 0.30

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30 [Intermediate] • Target: Joint Probability and Independence

Given $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cup B) = 0.7$, find $P(A \cap B)$.

a. 0.20

b. 0.10

c. 0.30

d. 0.15

31 [Intermediate] • Target: Joint Probability and Independence

Are the events in question 2 statistically independent?

a. Yes, because $P(A)P(B) = 0.5 \times 0.4 = 0.20 = P(A \cap B)$

b. No, because $P(A) \neq P(B)$

c. No, because $P(A \cup B) \neq 1$

d. Cannot be determined without conditional probabilities

32 [Intermediate] • Target: Conditional Probability and the Chain Rule

Given $P(A) = 0.6$ and $P(A \cap B) = 0.18$, calculate the conditional probability $P(B|A)$.

a. 0.30

b. 0.108

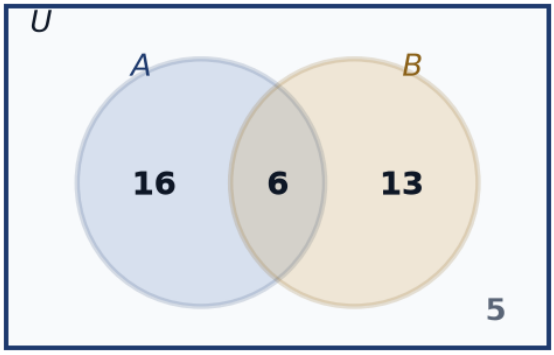
c. 0.42

d. 0.25

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33 [Intermediate] • Target: Conditional Probability and the Chain Rule

In an autoregressive language model, write the joint probability of a sequence of three tokens $P(w_1, w_2, w_3)$ using the probability Chain Rule.



a. $P(w_1)P(w_2|w_1)P(w_3|w_1, w_2)$

b. $P(w_1) + P(w_2) + P(w_3)$

c. $P(w_1)P(w_2)P(w_3)$

d. $P(w_3|w_2)P(w_2|w_1)$

34 [Intermediate] • Target: Conditional Probability and the Chain Rule

If $P(A|B) = 0.8$ and $P(B) = 0.25$, find $P(A \cap B)$.

a. 0.20

b. 0.55

c. 0.3125

d. 0.10

35 [Intermediate] • Target: Law of Total Probability

Factory line 1 produces 60% of microchips with defect rate 2%. Line 2 produces 40% with defect rate 5%. What is the total probability that a random chip is defective?

a. 0.032

b. 0.070

c. 0.035

d. 0.025

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36 [Intermediate] • Target: Law of Total Probability

In question 7, if a selected chip is defective, what is the probability it came from Line 2?

a. $\frac{0.020}{0.032} = 0.625$

b. 0.400

c. 0.375

d. 0.500

37 [Intermediate] • Target: Law of Total Probability

State the Law of Total Probability for an exhaustive partition $\{A_1, A_2, \ldots, A_k\}$ of the sample space.

a. $P(B) = \sum_{i=1}^{k} P(B|A_i)P(A_i)$

b. $P(B) = \prod_{i=1}^{k} P(B|A_i)$

c. $P(B) = \max_i P(B|A_i)$

d. $P(B) = \sum_{i=1}^{k} P(A_i|B)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

In medical screening, a disease has prevalence $P(D) = 0.01$. The test has sensitivity $P(+|D) = 0.99$ and false positive rate $P(+|\sim D) = 0.02$. What is the posterior probability $P(D|+)$ that a positive patient has the disease?

a. $\approx 33.3\%$

b. 99.0%

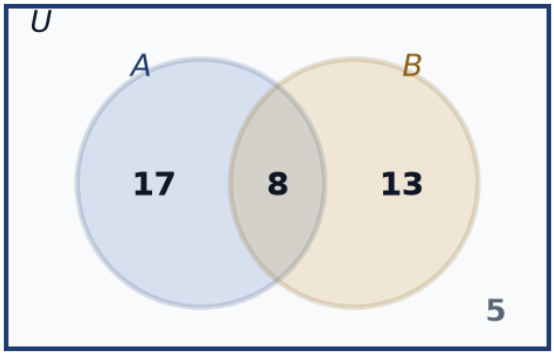
c. 50.0%

d. 5.0%

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39 [Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 1.01 Vectors and Embeddings in AI

What is the foundational simplifying assumption of the Naive Bayes text classifier?



a. Features (words) are conditionally independent given the class label

b. All prior probabilities are equal

c. The vocabulary size is infinite

d. Features follow a uniform distribution

40 [Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

In Maximum A Posteriori (MAP) estimation, how does the optimization objective differ from standard Maximum Likelihood Estimation (MLE)?

a. MAP incorporates a prior distribution: $\theta_{MAP} = \arg\max[\ln P(D|\theta) + \ln P(\theta)]$

b. MAP ignores all observed training data

c. MAP only applies to linear models

d. MAP minimizes variance instead of error

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41 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider a dataset of n data points, each residing in $\mathbb{R}^D$. We are performing Principal Component Analysis (PCA) to reduce the dimensionality. The first three principal components capture 70% of the total variance, the first six capture 85%, and the first ten capture 92%. You are tasked with selecting a subset of principal components for a downstream machine learning model. If the model requires that at least 90% of the original variance be retained, and you want to use the minimum possible number of principal components, what is the dimensionality of the transformed data?

a. 3

b. 6

c. 10

d. D

42 [Challenging] • Target: Joint Probability and Independence • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Consider a discrete random variable X with probability mass function (PMF) given by $P(X = k; \theta) = \frac{\theta^k e^{-\theta}}{k!}$ for $k = 0, 1, 2, \ldots$, where $\theta > 0$ is a parameter. Suppose we have an i.i.d. sample $x_1, x_2, \ldots, x_n$ from this distribution. We want to find the maximum likelihood estimate (MLE) for θ that maximizes the expected value of θ conditional on the sum of the observations, $S = \sum_{i=1}^{n} x_i$. Specifically, we are interested in the value of θ that maximizes $E[\theta | S = s]$. For a Poisson distribution, it is known that $E[\theta | S = s] = \frac{\theta s}{\theta + s}$ if θ itself were distributed as $Gamma(s + 1, 1)$. However, here θ is a fixed but unknown parameter. This question is about finding a parameter value that optimizes a related quantity. Let's consider a modified objective function for estimation that relates to the Fisher Information. Suppose we are interested in maximizing $L(\theta) = \frac{1}{n} \sum_{i=1}^{n} \log P(x_i; \theta) - \theta$. Find the value of θ that maximizes this modified likelihood function, given a sample where $n = 5$ and $\sum_{i=1}^{5} x_i = 15$.

a. 3

b. 5

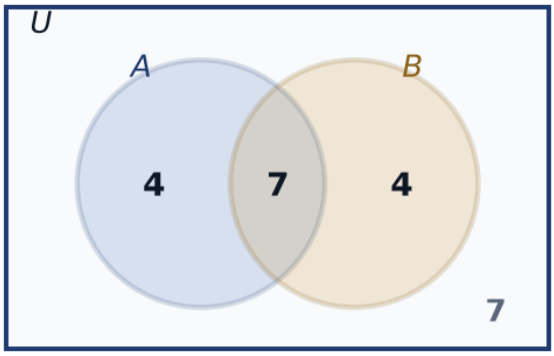
c. 15

d. 2

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43 [Challenging] • Target: Conditional Probability and the Chain Rule • Cross-Lesson: 6.01 Linear Regression and OLS

A national health organization is investigating the relationship between daily screen time (in hours) and the number of hours of sleep reported by adults. They collect data from a sample of 1200 adults. The data is summarized in a two-way table, where the row totals represent categories of daily screen time and the column totals represent categories of reported sleep hours. The organization is particularly interested in the cells where adults report less than 4 hours of screen time and get 7 or more hours of sleep. If the overall row total for 'less than 4 hours of screen time' is 250, and the overall column total for '7 or more hours of sleep' is 300, what is the expected count for the cell representing adults with less than 4 hours of screen time who get 7 or more hours of sleep, assuming independence between screen time and sleep hours?



a. 62.50

b. 75.00

c. 83.33

d. 250.00

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44 [Challenging] • Target: Law of Total Probability • Cross-Lesson: 7.01 Logistic Regression and Classification

A machine dispenses a liquid into containers. The amount of liquid dispensed, in milliliters, follows a continuous uniform distribution over the interval $[95, 105]$. A quality control inspector randomly selects 5 containers and measures the amount of liquid in each. Let X_i be the amount of liquid in the ith container, for $i = 1, 2, 3, 4, 5$. The company has implemented a new calibration process that, on average, increases the mean amount of liquid dispensed by δ milliliters, but also increases the variance by σ^2. The inspector is specifically looking for evidence that the calibration process has *decreased* the variability of the dispensed liquid while maintaining a satisfactory average fill. If the current calibration process yields a mean fill of $\mu_0 = 100$ ml and a standard deviation of $\sigma_0 = 2$ ml, and the new process is hypothesized to have a mean fill of $\mu_1 = 100 + \delta$ ml and a standard deviation of $\sigma_1 = 2 - \sigma_{dev}$ ml, where $\delta > 0$ and $\sigma_{dev} > 0$ are unknown positive constants, what is the minimum value of σ_{dev} such that the probability of falsely concluding that the new process has *increased* variability (i.e., Type I error) is at most 0.01, assuming the mean fill remains at 100 ml (i.e., $\delta = 0$)?

a. The probability cannot be determined without knowing the specific value of δ.

b. Approximately 1.01

c. Approximately 1.37

d. Approximately 2.00

45 [Challenging] • Target: Bayes Theorem in Machine Learning and Screening • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A manufacturing company is testing a new quality control process. They have collected data on the number of defects found in batches of a product, categorized by the production shift. The data is summarized in the two-way table below: | Shift | Defects Found | No Defects Found | Total | |---|---|---|---| | Morning | 15 | 85 | 100 | | Afternoon | 25 | 75 | 100 | | Evening | 10 | 90 | 100 | | **Total** | **50** | **250** | **300** | If a test of independence is to be performed on this data, what is the expected number of defects found in the afternoon shift?

a. 20.00

b. 16.67

c. 25.00

d. 8.33

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